

any matrix with
the $P(n, k)$
pattern



one recurrence,
order $2k + 2$



$\text{null } A =$
 $\dim \ker(T - I)$
 $\leq 2k + 2$



$= 2k + 2$
 $\Leftrightarrow T = I$

the ceiling equals the upper bound: no slack either way